Datasets for Training a Zero-Shot Voice Cloning Model

Survey of open speech corpora, licensing, merging strategy, and a data-cleaning pipeline | October 2026

1. Summary

A zero-shot voice cloning model learns to imitate an unseen speaker from a short reference clip. The single most important property of its training data is **speaker diversity** (thousands of distinct voices), followed by clean audio and accurate, punctuated transcripts. The standard approach is two-tiered: pretrain on a very large, moderately noisy corpus to learn general voice and prosody variation, then fine-tune on a smaller, very clean corpus to raise audio quality.

Recommended starting points:

- **Pretraining:** Emilia / Emilia-Large (multilingual, spontaneous), HiFiTTS-2 and Libriheavy (English, read speech).
- **Fine-tuning:** LibriTTS-R, VCTK, EARS, and Expresso for clean and expressive speech.
- **Indian languages:** IndicVoices-R combined with IndicTTS.
- **Licensing:** several high-value sets are non-commercial. Track the license of every clip from day one.

2. Tier 1: Large-Scale Pretraining Data

These corpora provide the speaker diversity that zero-shot generalisation depends on. Most are derived from audiobooks or in-the-wild web audio and need filtering before use.

Dataset	Scale	Description	License
Emilia	101k h; En, Zh, De, Fr, Ja, Ko; 24 kHz	In-the-wild speech from podcasts and video; mostly spontaneous, wide range of styles.	CC BY-NC 4.0
Emilia-YODAS	~114k h; multilingual	Extension that, together with Emilia, forms Emilia-Large (~216k h).	CC BY 4.0
HiFiTTS-2 (NVIDIA, 2025)	36.7k h @ 22.05 kHz (5k spk); 31.7k h @ 44.1 kHz (4.6k spk)	LibriVox-derived, bandwidth-filtered, with rich quality metadata. Repo holds manifests; audio is fetched from LibriVox.	CC BY 4.0
Libriheavy	50k h English	Read speech with punctuation, casing, and text context, which helps TTS prosody.	CC BY 4.0
MLS (Multilingual LibriSpeech)	44.5k h English + ~6k h in 7 other languages	Read audiobook speech. Transcripts lack punctuation, so restore it before TTS use.	CC BY 4.0
WenetSpeech4TTS	12.8k h Mandarin	Refined from WenetSpeech, with quality-tiered subsets (e.g. Premium).	Non-commercial; gated
GigaSpeech	~10k h English	Podcasts, YouTube, and audiobooks; multi-domain.	Usage agreement; verify terms

3. Tier 2: Clean Fine-Tuning Data

Use these to raise output fidelity and expressiveness after pretraining. Even 'clean' sets benefit from quality filtering. For example, LibriTTS-R contains some samples that were distorted by the restoration process.

Dataset	Scale	Description	License
LibriTTS-R	585 h; 2,456 speakers; 24 kHz	LibriTTS with speech restoration applied; same utterances, much better sound quality.	CC BY 4.0
VCTK 0.92	~44 h; 110 speakers; 48 kHz	Studio recordings, many English accents. Speaker p315 has no transcripts and p280 is missing mic2 audio.	CC BY 4.0
EARS (Meta)	100 h; 107 speakers; 48 kHz	Anechoic recordings; 7 reading styles, 22 emotions, whisper to yelling, freeform speech.	Verify (believed CC BY-NC)
Expresso (Meta)	47 h; 4 speakers; 48 kHz	Highly expressive read and improvised speech in 26 styles.	CC BY-NC 4.0

4. Indian-Language Data

Dataset	Scale	Description	License
IndicVoices-R (AI4Bharat)	1,704 h; 10,496 speakers; 22 languages (9-175 h each)	Enhanced from ASR data to TTS quality; includes a zero-shot speaker benchmark. Fine-tuning an English-pretrained model on IndicTTS + IV-R improved zero-shot speaker generalisation.	Gated; check terms
IndicTTS (IIT Madras)	Studio quality; few speakers per language	Clean, consistent recordings that work well as a quality anchor alongside IV-R.	Check terms

5. Merging Strategy and Cleaning Pipeline

Critical pitfall: overlapping sources

LibriTTS, LibriTTS-R, Libriheavy, HiFiTTS-2, and MLS-English are all derived from LibriVox. They share speakers and even identical recordings. Without deduplication you will over-weight some voices and, worse, leak evaluation speakers into training. Deduplicate by LibriVox speaker ID and book/chapter ID, and remove LibriSpeech test-clean speakers if you plan to report on that standard zero-shot benchmark.

Recommended processing steps

1. **Ingest new raw audio with Emilia-Pipe.** It is open source and handles source separation, speaker diarization, VAD segmentation, ASR, and filtering for in-the-wild data.
2. **Standardise the format.** Resample everything to one rate (24 kHz is common), convert to mono, apply loudness normalisation (around -23 LUFS), and trim leading and trailing silence.
3. **Filter by quality.** Score clips with DNSMOS or UTMOS and drop the low scorers. Estimate effective bandwidth and drop audio that was upsampled from low-bitrate sources.
4. **Verify transcripts.** Re-transcribe with Whisper and discard clips whose WER against the provided transcript exceeds a threshold. Restore punctuation and casing where missing.
5. **Check speaker purity.** Use speaker embeddings (ECAPA-TDNN or WavLM) to reject clips containing more than one speaker.
6. **Segment.** Keep utterances of roughly 3-20 seconds with aligned, punctuated text.
7. **Track provenance.** Store source dataset, license, and speaker ID for every clip in your manifest so a commercial-safe subset can be exported at any time.

6. Suggested Dataset Mixes

Goal	Pretraining	Fine-tuning
Research, English-first	Emilia-Large + Libriheavy	LibriTTS-R + VCTK + EARS (+ Expresso)
Commercial-safe	Emilia-YODAS + HiFiTTS-2 + Libriheavy + MLS	LibriTTS-R + VCTK
Multilingual incl. Indian	Emilia-Large + MLS + IndicVoices-R	IndicTTS + LibriTTS-R + VCTK

Scale guidance. A true foundation model generally needs tens of thousands of hours drawn from thousands of speakers. Fine-tuning an existing open zero-shot TTS model (for example, one pretrained on Emilia) can work with a few hundred to a few thousand clean, filtered hours.

7. Responsible Use

Zero-shot cloning can be misused for impersonation and fraud. Building in reference-speaker consent checks and audio watermarking from the start is good practice, and commercial deployments are increasingly expected to have both. Respect each dataset's license, and only add your own scraped audio when you hold the rights to it.

Appendix: Download Links

Dataset	Link	Notes
Emilia / Emilia-Large	https://huggingface.co/datasets/amphion/Emilia-Dataset	Gated (agree to share contact info); supports streaming
Emilia-Pipe	https://github.com/open-mmlab/Amphion/tree/main/preprocessors/Emilia	Cleaning pipeline for your own audio
HiFiTTS-2	https://huggingface.co/datasets/nvidia/hifitts-2	Manifests only; audio downloaded from LibriVox at 48 kHz
Libriheavy	https://github.com/k2-fsa/libriheavy	Scripts and transcripts; audio from Libri-Light
MLS	https://openslr.org/94/	English is 2.4 TB FLAC or 651 GB Opus
WenetSpeech4TTS	https://huggingface.co/datasets/Wenetspeech4TTS/WenetSpeech4TTS	Gated; non-commercial
GigaSpeech	https://github.com/SpeechColab/GigaSpeech	Access request required
LibriTTS-R	https://www.openslr.org/141/	CC BY 4.0
LibriTTS (original)	https://www.openslr.org/60	CC BY 4.0
VCTK 0.92	https://datashare.ed.ac.uk/handle/10283/3443	Direct zip: VCTK-Corpus-0.92.zip
EARS	https://github.com/facebookresearch/ears_dataset	48 kHz anechoic
Expresso	https://speechbot.github.io/expresso/	CC BY-NC 4.0
IndicVoices-R	https://huggingface.co/datasets/ai4bharat/indicvoices_r	Also per-language tars: GitHub
IndicTTS	https://www.iitm.ac.in/donlab/indictts/	Review license terms

Licenses and access terms can change. Confirm each dataset's current terms on its official page before training, especially for commercial use. This report is not legal advice.